

ECE 447

Lesson 38

Error Correction: Parity, CRC, and Hamming Codes



UNITED STATES
AIR FORCE
ACADEMY

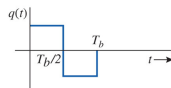
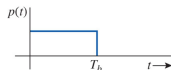
SCHEDULE AND ADMIN

- **Schedule.**
 - Lesson 38 - Error correction
 - Lesson 39 - SDR Lab 8: Capstone image transfer
 - Lesson 40 - Advanced topics: OFDM, MIMO, CDMA
 - Lesson 41 - Course review
- **Admin**
 - **HW8.** Due Lesson 39 (Thu, 3 Dec) to Gradescope.

REVIEW

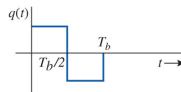
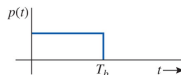
Matched Filters and Binary Signaling Performance

- Why?
- General binary case for real-valued pulses: $h(t) = p(T_b - t) - q(T_b - t)$
- $P_e = P_b = Q\left(\sqrt{\frac{E_p + E_q - 2E_{pq}}{2N_0}}\right)$, where $E_{pq} = \int_0^{T_b} p(t)q(t)dt$ is the inner product or correlation of the two pulses
- Example problem: Find the matched filter impulse response $h(t)$ and calculate the total bit error probability P_e assuming equiprobable transmission of 0 or 1



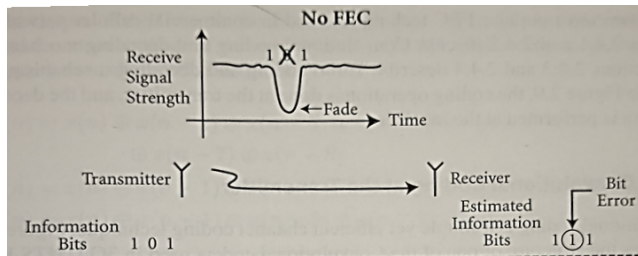
REVIEW

- Example problem: Find the BER for a system using **polar signaling** pulses $\pm 3p(t)$ with $T_b = 1\mu s$ and $N_0 = \frac{25\mu W}{Hz}$.



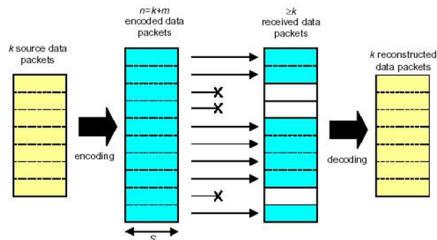
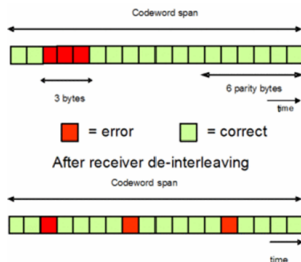
ERROR CORRECTION

- Bit errors occur for a variety of reasons, but can correct many by pre-processing bits at the transmitter - **forward error correction (FEC)**
- "Error correction coding requires a strong mathematical background" - so true!
- Types of codes (in order of complexity): repetition, block, convolutional, [Viterbi algorithm](#), turbo, low-density parity check (LDPC), polar



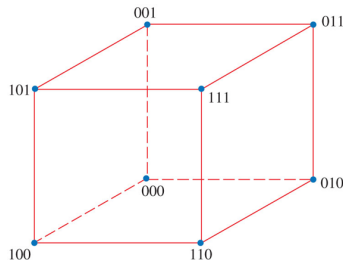
INTERLEAVING

- First, though - errors often consecutive in time and/or frequency
- Interleaving spreads out code symbol errors at receiver - decoder receives few to zero consecutive errors
- Synchronization required!



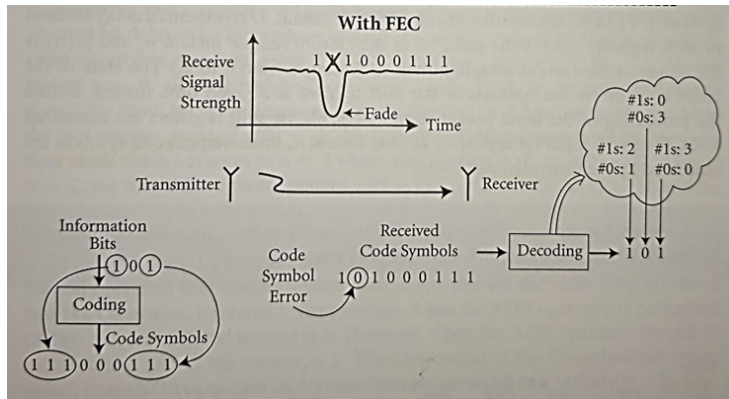
ERROR CORRECTION BASICS AND TERMINOLOGY

- **Code rate:** ratio of data bits k to total bits n - creates an (n, k) code
- d is bit vector **message** (data)
- c is bit vector **codeword**
- Number of redundant (aka check aka parity) bits is $m = n - k$
- **Hamming distance:** number of bits that are different between two bit vectors
- **Hamming space:** multi-dimensional cube; spheres drawn around possible codeword vertices - the radius spheres can have without overlapping equals the number of errors that can be corrected
- To *correct* t errors, min Hamming distance $d_{min} = 2t + 1$
- To *detect* t errors, min Hamming distance $d_{min} = t + 1$



REPETITION CODE

- Repeat each bit a certain number of times, then transmit
- To decode at receiver, use a simple majority rule



HAMMING CODE

- Type of linear block code
- $c = dG$, where G is called the **generator matrix**
- We transmit c and receive r , which may have errors
- $s = rH^T$ is called the **syndrome**, where H is the **parity check matrix**
- The row in H^T that matches s is the bit of the codeword in error - flip that bit
- If $s = 0$, then no errors
- **Coding gain:** reduction in req'd SNR to achieve specific BER

WHAT DO WE "GAIN"?

- **Coding gain:** reduction in req'd SNR to achieve specific BER

